

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) I Semester Examination, 2016-17

Subject: Computer Science

**Paper: CS-101 (Problem Solving through C Programming) /
BCS-101 (Introduction to Computer Programming Through C)**

Time: Three hours

Maximum Marks: 70

Instructions: Answer any five questions, including Question No. 1, which is compulsory. All undefined symbols have their usual meanings.

Question 1. Attempt all questions:

[2 × 9 = 18]

- (a) Define Algorithm. What are its characteristics?
- (b) What is the difference between '=' operator and '==' operator?
- (c) When is the register storage class most useful? Give a suitable example.
- (d) What is the role of a Preprocessor in 'C' programming?
- (e) Why is the '&' operator not used with array name in a `scanf()` statement?
- (f) What do you mean by '→' operator in C? When is it used?
- (g) What will be the output of the following code?

```
void main()
{
    int a=3, b=4;
    b %= 3+4;
    a *= a+5;
    printf("%d_%d", b, a);
}
```

- (h) Consider the following code segment and write its final output:

```
void main()
{
    int x=3, z;
    z = x / ++x;
    printf("x=%d_z=%d", x, z);
}
```

- (i) What is the output of the following code?

```
void main()
{
    int a=30, b=0, c;
    c = (a != 10) && (b = 0);
    printf("c=_%d", c);
}
```

Question 2.

- (a) What is the difference between top-down approach and bottom-up approach of programming? [2]
- (b) Draw a flowchart to find the roots of a quadratic equation $ax^2 + bx + c = 0$. [4]
- (c) List the rules for naming a variable in C. [2]
- (d) Explain the difference between nested if-else clause and else if clause used in C language. [5]

Question 3.

- (a) Define function. What do you mean by function prototype? Is it always compulsory to write function prototype statement while using a function in a 'C' program? Justify your answer with one example. [5]
- (b) Describe the two ways of passing arguments to function? When do you prefer to use each of them? [4]
- (c) Write a program using function to find m^n where $m, n \geq 0$. [4]

Question 4.

- (a) State the difference between automatic and static storage class variables by using appropriate example. [4]
- (b) Write your own function for the following operations on string: [6]
 - (i) Comparison of two strings
 - (ii) Copy one string into another string
- (c) What is the use of enumerated data type in 'C' programming? [3]

Question 5.

- (a) How are 2D array elements stored in computer memory in 'C' programming? [2]
- (b) Write a 'C' program to find out transpose of a given matrix. [4]
- (c) What is meant by the following terms? Give examples of use of each of them. [7]
 - (i) Array of pointers
 - (ii) Pointer to Array

Question 6.

- (a) Explain the difference between a 'Structure' and a 'Union' with suitable examples. [4]
- (b) How do you pass parameters to main() function in a 'C' program? Explain with a suitable example. [4]

- (c) Can a structure be used within another structure? Give appropriate example to support your answer. [5]

Question 7.

- (a) What is the need of dynamic memory allocation? Explain the role of `realloc()` function with its proper syntax. [5]
- (b) Define Macro. In what situation is it better to use Macro instead of using Function? [3]
- (c) What is conditional compilation? How does it help a programmer? Explain with an example. [5]